

Aaron Chou

☎ +886-9-70819587 | ✉ aaron890707@gmail.com | 🏠 uuuloon.github.io | 🌐 Aaron Chou | 🌀 uuuloon

EDUCATION

National Taiwan University (NTU) , <i>B.Sc. in Mechanical Engineering</i>	Taipei, Taiwan
• Last 60 GPA: 4.03/4.3 CGPA: 3.80/4.3	Sept. 2018 – Jan. 2023
• Coursework: Automatic Control, Digital Control System, Kinematics, Dynamics, Computer Programming	
Aoyama Gakuin University (AGU) , <i>Exchange Program</i>	Tokyo, Japan
• Coursework: Data Structures and Algorithms, Introduction to Computer Systems	Sept. 2022 – Jan. 2023

PUBLICATIONS

[1] Y.L. Chou, Y.S. Luo, L.C. Wang, H. Mandala, G.H.G. Christmann, C.Y. Lee, W.C. Chen, et al. *Feasibility-Guided Planning over Multi-Specialized Locomotion Policies*, IEEE International Conference on Robotics and Automation (ICRA), 2026. ([link](#))

EXPERIENCE

Inventec Corporation , AI Center, Robotics, Advisor: Dr. Wei-Chao Chen	Taipei, Taiwan
<i>Robotics Research Engineer — Quadruped Robots</i>	Mar. 2025 – Present
• Vision-Language Navigation (VLN) : Research VLN (link) in unseen environment through natural language instruction.	
• Learning-Based Control : Developed a low-level control SDK for quadruped platforms with no official API, enabling deployment (link) of learning-based locomotion policies trained in IsaacGym and establishing reusable infrastructure.	
• Terrain-Aware Navigation : Trained a 15 cm gap-traversal policy, implemented a feasibility-guided planning framework supporting skill selection on hybrid terrain, and contributed to an ICRA 2026 submission [1].	
• SLAM Optimization : Improved single-LiDAR SLAM by dual-LiDAR fusion (link) in ICRA 2025 QRC, achieving a 65% reduction in mapping-to-navigation time and enabling rapid terrain reconstruction in dynamic environments.	
Flytech Technology Co. Ltd.	Taipei, Taiwan
<i>Robotics Engineer — Autonomous Mobile Robots</i>	Mar. 2024 – Jan. 2025
• Engineered a precise docking system (demo) by integrating motion control, localization, and a customized rack tracker, achieving navigation accuracy of $\pm 1-1.5$ cm to minimize redocking attempts in production deployments.	
• Developed and maintained 10+ ROS packages and delivered a patrol demo showcased at Computex 2024.	
Chien Kuo High School , Robotics team, FRC#8020 Cyberpunk	Taipei, Taiwan
<i>Youth Mentor — Mechanical Design</i>	Feb. 2022 – Jul. 2023
• Mentored 30+ students for robot design, CAD/CAM, CNC operation, and testing of the competition-ready robot (demo).	

PROJECTS

Isaac_MoveIt Manipulator Integration , <i>Collaborative Project</i> (demo)	Jun. 2025 – Present
• Integrated IsaacSim with MoveIt2 , enabling manipulator control in simulation for future dexterous research.	
Differential-Wheeled Robot , <i>Independent Project</i>	Mar. 2022 – Aug. 2022
• Developed a SLAM-capable mobile robot by integrating LiDAR with an edge-computing platform for real-time mapping with affordable cost under 120\$, laying the groundwork for later AMR development and quadruped research.	

SKILLS

Robotics: IsaacGym, IsaacSim, ROS/ROS2 (navigation2, MoveIt2, FAST_LIO), Path Planning, Sensor Fusion, SLAM
Toolkits: Docker, Git, Blender, PCL, SolidWorks/CAM, LaTeX
Programming: Python, C++, Shell Scripting (Bash)

HONORS & COMPETITIONS

2025 ICRA Quadruped Robot Challenge (QRC) — Participation Certificate, Autonomous (demo)	Atlanta, GA
2023 Admission to Tohoku University , M.S. in Mechanical Engineering	Tohoku, Japan
2022 FIRST Robotics Competition (FRC) Sacramento Regional, Finalist (Team Mentor)	Sacramento, CA
2020 Fall Dean's List Award (top 5% of the class in the semester)	Taipei, Taiwan